

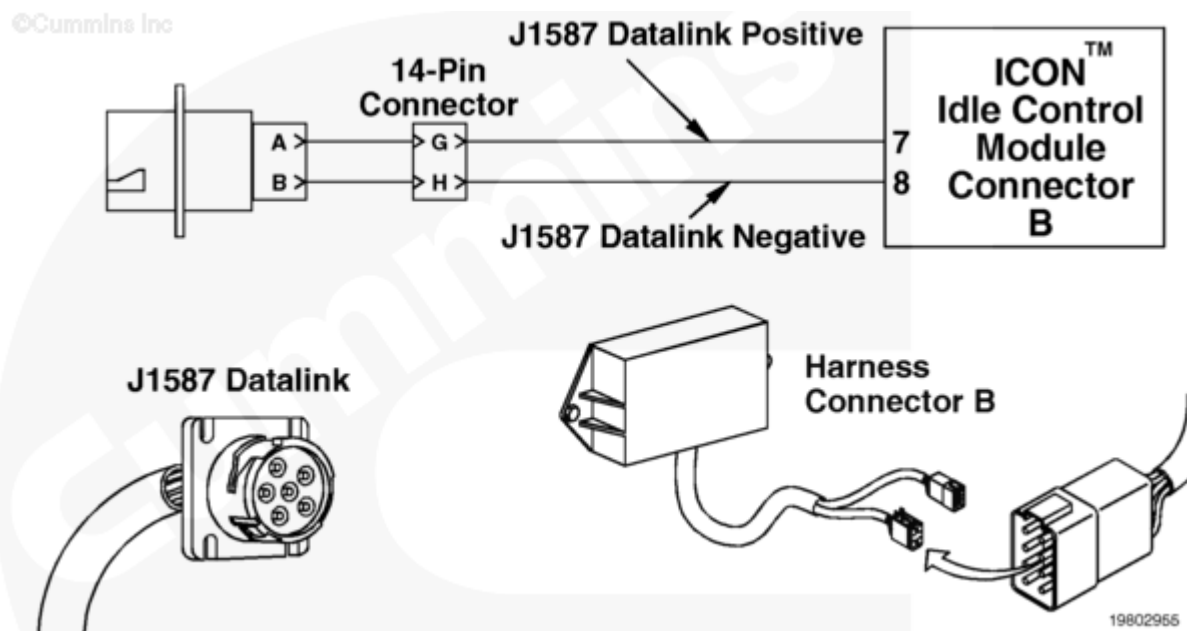
Fault Code: 211 (Aftermarket and OEM)

Additional OEM or Vehicle Diagnostic Codes Have Been Logged

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 211 PID(P): SPN: FMI: Lamp: None SRT:	Additional OEM or Vehicle Diagnostic Codes Have Been Logged. Some information expected on the J1587 datalink by the ICON™ idle control module was not received from the engine ECM.	The ICON™ system can be disabled and mandatory shutdown can be disabled. Engine will start normally.



Circuit Description

The J1587 datalink provides communication between the engine electronic control module (ECM) and the ICON™ idle control module. The above circuit diagram can vary, such as connector or pins, depending on the vehicle make or model. OEM installations can possibly provide the harnessing between the idle control module and other ICON™ devices.

Component Location

The J1587 datalink is located in the OEM wiring harness. The ICON™ idle control module typically connects into the J1587 wiring harness behind the in-cab datalink service connector. The ICON™ module can be located in a different location depending on the vehicle application.

Shoptalk

This fault is set when information that is required by the ICON™ idle control module is **not** received from the engine ECM. It is also set when the ICON™ idle control module is unable to set the idle speed correctly. Check the engine ECM for active faults and troubleshoot those first.

Module communications are listed below.

Engine ECM to ICON™ idle control module:

- RPM
- Coolant temperature
- Oil temperature
- Throttle position
- Clutch switch
- Cruise switch
- Vehicle speed parameters.

ICON™ idle control module to engine ECM:

- speed control message parameter once per second.

The ICON™ system can display **only** the present active fault code. If more than one fault code is active at the same time, the ICON™ system flashes out the highest priority fault. After the fault has been corrected then the next active fault will be flashed out.

Note: The ICON™ electronic service tool can display more than one active and or inactive fault codes at the same time.

Warnings and Cautions

CAUTION

To reduce the possibility of damaging a new ICON™ idle control module, all other active fault codes must be investigated prior to replacing the ICON™ idle control module.

CAUTION

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement:

Part Number 3822758 - male Deutsch/AMP/Metri-Pack test lead

Part Number 3822917 - female Deutsch/AMP/Metri-Pack test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Read the fault codes.	
	STEP 1A. Use the fault flashout feature or the ICON™ electronic service tool to read the fault code.	Fault Code 211 inactive
STEP 2.	Check the ICON™ engine harness.	
	STEP 2A. Check for a short circuit in the J1587 datalink.	Pin A: 2.5 to 5 VDC, Pin B: 0 to 2.5 VDC, Pin C: 10 to 27 VDC, Pin E: 0 VDC
	STEP 2B. Check the ICON™ engine harness and the ICON™ idle control module connector pins.	No damaged pins
	STEP 2C. Check the complete harness for a short circuit from pin to pin.	More than 100k ohms
	STEP 2C-1. Identify whether the ICON™ system is an Aftermarket or an OEM.	ICON™ system is Aftermarket
	STEP 2C-2. Check the cab harness for a short circuit from pin to pin.	More than 100k ohms
	STEP 2D. Check the complete harness for a short circuit to ground.	More than 100k ohms
	STEP 2D-1. Identify whether the ICON™ system is an Aftermarket or an OEM.	ICON™ system is Aftermarket
	STEP 2D-2. Check the cab harness for a short to ground.	More than 100k ohms
STEP 3.	Clear the fault code.	

STEPS		SPECIFICATIONS
	STEP 3A. Disable the fault code.	Fault Code 211 cleared

STEP 1. Read the fault codes.

STEP 1A. Use the fault flashout feature or the ICON™ electronic service tool to read the fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect the ICON™ electronic service tool.

Action	Specification/Repair	Next Step
• Initiate the datalink test button on the service tool to determine if the datalink is working properly.	Fault Code 211 inactive Refer to Inactive or Intermittent Fault Codes, Procedure 019-362 (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html).	3A
		2A

STEP 2. Check the ICON™ engine harness.

STEP 2A. Check for a short circuit in the J1587 datalink.

Conditions :

- Disconnect the ICON™ idle control module B connector from the ICON™ idle control module.
- Make sure the 14-pin pass-through connector is connected.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
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• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Wire insulation damage • Moisture in or on the connector • Missing or damaged connector seals • Connector shell broken • Dirt or debris in or on the connector pins. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	2C
	Repair the damaged pins. • Flush the dirt, debris, or moisture from the connector pins using electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Repair the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html). • Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html). • Replace the ICON™ idle control module. Refer to Procedure 019-358 (/qs3/pubsys2/xml/en/procedures/97/97-019-358.html). • Repair or replace the OEM wiring harness as necessary.	3A

STEP 2C. Check the complete harness for a short circuit from pin to pin.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ idle control module B connector from the ICON™ idle control module.

Action	Specification/Repair	Next Step
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• Measure the resistance from pin 7 in the ICON™ idle control module B harness connector to all other pins in the connector. • Measure the resistance from pin 8 in the ICON™ idle control module B harness connector to all other pins in the connector. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms	2D
		2C-1

STEP 2C-1. Identify whether the ICON™ system is an Aftermarket or an OEM.

Conditions : None		
Action	Specification/Repair	Next Step
Refer to Procedure 209-017 (/qs3/pubsys2/xml/en/procedures/97/97-209-017.html).	ICON™ system is Aftermarket	2C-2
	Check the OEM wiring harness for a short circuit pin to pin on the J1587 datalink circuit.	3A

STEP 2C-2. Check the cab harness for a short circuit from pin to pin.

Conditions :

- Turn keyswitch OFF.
- Disconnect the 14-pin pass-through connector at the vehicle bulkhead.

Action	Specification/Repair	Next Step
• Measure the resistance from pin G in the 14-pin pass-through connector, cab harness side, to all other pins in the connector, except pin H. • Measure the resistance from pin H in the 14-pin pass-through connector, cab harness side, to all other pins in the connector, except pin G. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms Repair or replace the ICON™ engine harness • Repair the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html) or 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html). • Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).	3A
	Repair or replace the cab harness • Repair the cab harness. Refer to Procedure 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html). • Replace the cab harness. Refer to Procedure 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).	3A

STEP 2D. Check the complete harness for a short circuit to ground.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ idle control module B connector from the ICON™ idle control module.

Action	Specification/Repair	Next Step
• Measure the resistance from pin 7 in the ICON™ idle control module B harness connector to engine block ground. • Measure the resistance from pin 8 in the ICON™ idle control module B harness connector to engine block ground. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms Replace the ICON™ idle control module. Refer to Procedure 019-358 (/qs3/pubsys2/xml/en/procedures/97/97-019-358.html).	3A
		2D-1

STEP 2D-1. Identify whether the ICON™ system is an Aftermarket or an OEM.**Conditions :**

None

Action	Specification/Repair	Next Step
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Refer to Procedure 209-017 (/qs3/pubsys2/xml/en/procedures/97/97-209-017.html).	ICON™ system is Aftermarket	2D-2
	Check the OEM wiring harness a short circuit to ground on the J1587 datalink circuit.	3A

STEP 2D-2. Check the cab harness for a short circuit to ground.

Conditions : • Turn keyswitch OFF. • Disconnect the 14-pin pass-through connector at the vehicle bulkhead.		
Action	Specification/Repair	Next Step

• Measure the resistance from pin G in the 14-pin pass-through connector, cab harness side, to engine block ground. • Measure the resistance from pin H in the 14-pin pass-through connector, cab harness side, to engine block ground. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms Repair or replace the ICON™ engine harness • Repair the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html) or 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html). • Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).	3A
	Repair or replace the cab harness • Repair the cab harness. Refer to Procedure 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html). • Replace the cab harness. Refer to Procedure 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).	3A

STEP 3. Clear the fault code.

STEP 3A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
• Cycle the keyswitch to verify the fault code is inactive.	Fault Code 211 cleared	Repair complete
	Troubleshoot any remaining active fault codes.	Appropriate troubleshooting charts

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